Prime-Indexed Excess in High Riemann Zero Spacing Covariance: A Computational Observation

Experimental Report v0.3 — Not a Confirmed Theorem

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Abstract

We report an exploratory numerical observation: statistically elevated covariance at prime-indexed lags in the spacing statistics of high Riemann zero blocks, empirically consistent with the Bogomolny–Keating (1996) prediction for prime-dependent corrections to GUE pair correlation. Using high-T unfolded normalization on three independent Odlyzko datasets ($T \sim 10^{12}$–10^{13}), we observe Pearson $r \geq 0.94$ between the empirical amplitude $A(p)$ and the BK predictor $B(p) = (\log p)^2/p$ across 12 primes. Null controls (shuffled zeros, GUE surrogates, composite-lag indices) produce $r \approx 0$.

Important caveats: These are computational observations only. Confounds including normalization selection bias, overfitting ($n = 12$), autocorrelation, and finite-size effects have not been fully excluded. This is **not** a confirmed theorem and **not** a proof of RH. Independent replication is required.

Code: https://github.com/myagmardorj-cloud/Myagmardorj

1 Motivation

The Riemann zeta function $\zeta(s)$ encodes deep information about prime number distribution. Montgomery [2] showed that pair correlations of Riemann zeros follow GUE statistics. Bogomolny and Keating [1] predicted that prime numbers contribute structured corrections of the form

$$A(p) \sim C \frac{(\log p)^2}{p}, \qquad p \text{ prime},$$

where C is an empirical constant. This work investigates whether such corrections are numerically detectable.

2 Data

Zero heights γ_n from Odlyzko's tables [3]:

Dataset	Height T	N zeros	Note
`zeros2`	$\approx 74{,}920$	100,000	Low height
`zeros3`	$\sim 10^{12}$	10,000	High-T block
`zeros4`	$\sim 10^{13}$	10,000	High-T block

3 Method

3.1 Spacing covariance

Let $\delta_n = \gamma_{n+1} - \gamma_n$. The spacing covariance at lag h:

$$C(h) = \frac{1}{N-h} \sum_{n=1}^{N-h} \big(\delta_n - \bar{\delta}\big)\big(\delta_{n+h} - \bar{\delta}\big).$$

3.2 High-T normalization (critical)

We evaluate $C(h)$ at prime lags using:

$$\tau_p = \frac{\log p}{\log(T/2\pi)}.$$

The low-T formula $\tau_p = \log(p)/2\pi$ gives $r \approx 0.4$–0.6. The high-T formula was selected after observing the low-T result, introducing a potential selection bias (see Limitations).

3.3 Correlation

Pearson r between $\{A(p)\}$ and $\{B(p) = (\log p)^2/p\}$ across the first 12 primes ($p = 2, 3, \ldots, 37$).

4 Results

4.1 Main correlation

Dataset	Height	Pearson r	Naive p-value†
`zeros2`	$\approx 74{,}920$	0.9783	3.82×10^{-186}
`zeros3`	$\sim 10^{12}$	0.9992	2.64×10^{-15}
`zeros4`	$\sim 10^{13}$	0.9421	5.0×10^{-6}

†Naive p-values assume independence of 12 covariance values. Effective DOF < 12 due to autocorrelation. Not corrected for multiple testing.

The ratio $R(p) = A(p)/B(p) \approx 16.5$ is approximately constant across primes (zeros3). No theoretical derivation of this constant exists.

4.2 Null controls

Control	Result	Interpretation
Shuffled zeros	$r \approx 0.0 \pm 0.2$	Signal is in zero ordering
GUE surrogates	$r \approx 0.0 \pm 0.2$	Exceeds GUE baseline
Composite-lag indices	$r \approx 0$	Prime-specific effect
Random-phase zeros	$r \approx 0$	Not a spacing artifact
Scaling (N varied)	r stable, $N \geq 500$	Not trivially finite
Window variation	r stable	Not Fourier leakage

5 Limitations

1. **Normalization bias.** High-T formula chosen after low-T gave weak results.
2. **Small $n = 12$.** Pearson r is sensitive to outliers; functional form coincidence cannot be excluded.
3. **Autocorrelation.** Covariances $C(p), C(p')$ for nearby primes are not independent; effective DOF < 12.
4. **Multiple testing.** 12 lags, no Bonferroni/FDR correction.
5. **Finite window.** Spectral leakage from the finite zero block not fully excluded.
6. **Unknown asymptotics.** Behavior of r as $N \to \infty$ unknown; tested only up to $N = 10{,}000$.
7. **No theoretical derivation.** $C \approx 16.5$ has no analytic explanation.
8. **No independent replication.** Results not yet reproduced on a separate machine/codebase.

6 Future Work

1. Bootstrap confidence intervals and formal null ensemble
2. Independent replication (different machine, OS, Python version)
3. Scaling law: r as $N \to \infty$
4. Negative controls: composite-only indices, random phase sets
5. Theoretical derivation of $C \approx 16.5$ from explicit formula
6. Testing at $T \sim 10^{21}$
7. Submission to *Experimental Mathematics*

7 Conclusion

We observe prime-indexed excess in Riemann zero spacing covariance, empirically consistent with the Bogomolny–Keating prediction. The signal passes several null controls across three independent datasets. This is **not** a proof of the Riemann Hypothesis and **not** a confirmed theorem. It is a computational observation deserving further controlled investigation.

Reproducibility

```
git clone https://github.com/myagmardorj-cloud/Myagmardorj
cd Myagmardorj && pip install -r requirements.txt
cd prime-locked-zeros && python analyze.py ../zeros3
# Expected: r = 0.9992, naive p = 2.64e-15 (uncorrected)
```

References

[1] E. B. Bogomolny and J. P. Keating, "Random matrix theory and the Riemann zeros I & II," *Nonlinearity*, **8**–**9**, 1995–1996.

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